

GAUSSIAN CHANNELS AND TRANSCENDENTAL BERNOULLI CONVOLUTIONS

GPT-6 ASTRA

ABSTRACT. We propose a Gaussian-channel proof of Varjú’s full-dimension theorem for transcendental Bernoulli convolutions. The analytic argument uses ordered conditional laws, randomly shifted grids, and disjoint blocks of independent digits. These give the weak algebraic approximation needed for the theorem directly, without the strong approximation theorem of Breuillard–Varjú. A form of Dimitrov’s bounded-order Taylor argument supplies the final arithmetic exclusion. The entropy–Mahler argument of Breuillard–Varjú is retained and reproduced. The remaining external inputs are exact dimensionality and classical polynomial root separation.

This is an unedited model output that led to the paper [KK26]. Prompting was done by C.K. and the prompter’s local context was used.

1. THE THEOREM AND THE ARGUMENT

Let $\varepsilon_0, \varepsilon_1, \dots$ be independent fair $\{0, 1\}$ -valued variables and put

$$X_\lambda = \sum_{j \geq 0} \varepsilon_j \lambda^j, \quad B_n(\lambda) = \sum_{j=0}^{n-1} \varepsilon_j \lambda^j, \quad \nu_\lambda = \mathcal{L}(X_\lambda).$$

Using signs $\{-1, 1\}$ instead changes the measure by an invertible affine map. All logarithms in this paper are natural.

Theorem 1.1 (Varjú [Var19a]). *If $\lambda \in (1/2, 1)$ is transcendental, then $\dim \nu_\lambda = 1$.*

Here is the proposed route. Write $\chi = \log(1/\lambda)$ and suppose $\alpha = \dim \nu_\lambda < 1$. Gaussian smoothing then has a definite derivative at every scale. This property passes to every convolution factor of ν_λ . An ordered binary split detects conditional variance, and the information chain rule adds these detections. Randomly shifting the grid controls the loss from splits very close to cell boundaries.

The resulting variance estimate is used directly. We place disjoint blocks of Bernoulli digits so that their variance lies in the same interval of physical scales. The total available information bounds their sum. For every $\chi < h < \log 2$, this yields

$$\overline{H}(B_n(\lambda); e^{-Cn}) < hn \quad \text{eventually, for every fixed } C > 0, \quad (1.1)$$

$$\overline{H}(B_n(\lambda); e^{-An \log n}) < hn \quad \text{infinitely often, for every fixed } A > 0. \quad (1.2)$$

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This is a proposed proof developed in a time-limited research session. The argument has been checked internally but has not been independently verified. Novelty of the proof has not been established.

The notation $\overline{H}$ denotes averaged quantization entropy, defined below. The first assertion gives small polynomial values at every large degree, the concentration conclusion of Hochman's theorem [Hoc14]. The second gives algebraic approximants with bounded Mahler measure. An elementary Taylor argument makes these two conclusions incompatible. The approximants need only have entropy bounded away from $\log 2$; they need not inherit the dimension deficit.

The theorem itself, the entropy–Mahler estimate, and the root-separation route to a common zero are established results [Var19a], [BV20], and [BV19]. The Taylor step is a version of Dimitrov's argument in [RV24, Lemma 4.6]. The candidate simplifications here are the ordered channel estimate and its direct use in the block-packing argument. The variance-summation method originates in Kittle [Kit25] and is developed for self-similar measures in Kittle–Kogler [KK25]. No claim that this general principle is new is intended. We include the details of the proposed argument so that these particular steps can be checked.

2. GAUSSIAN ROUGHNESS AND INFORMATION

We use translation-averaged entropy at scale as in [Var19b, Sections 1.2 and 2.2]. For bounded Y and $r > 0$, define

$$\overline{H}(Y; r) = \int_0^1 H(\lfloor Y/r + u \rfloor) du, \quad \mathcal{H}(Y; r) = h(Y + rZ) - h(rZ) = I(Y; Y + rZ),$$

where Z is an independent standard Gaussian. We write $\mathcal{H}(Y; t \mid R) = \mathcal{H}(Y; t) - \mathcal{H}(Y; R)$. Gaussian degradation and monotonicity of differential entropy under convolution give

$$0 \leq \mathcal{H}(Y; t \mid R) \leq \log(R/t) \quad (0 < t < R). \quad (2.1)$$

For independent bounded X, Y , put

$$D_{X,Y}(t, R) = \mathcal{H}(X + Y; t \mid R) - \mathcal{H}(X; t \mid R).$$

Conditioning on Y identifies this as

$$D_{X,Y}(t, R) = I(Y; X + Y + tZ) - I(Y; X + Y + RZ) \geq 0. \quad (2.2)$$

The two noises can be coupled by adding an independent Gaussian.

We use the standard comparison

$$\mathcal{H}(Y; r) = H(\lfloor Y/r \rfloor) + O(1) = \overline{H}(Y; r) + O(1). \quad (2.3)$$

The constants are absolute. Indeed, write $Y/r = K + U$, with $U \in [0, 1)$. The conditional channel information of U is bounded by Gaussian capacity, and $H(K \mid K + U + Z) = O(1)$ since $K - \lfloor K + U + Z \rfloor = -\lfloor U + Z \rfloor$ has uniformly bounded entropy. Two translated grids of the same mesh change entropy by at most $\log 2$. More generally, changing a mesh by a factor at most two changes entropy by at most $\log 3$ in the direction needed below.

Lemma 2.1. *If $\alpha = \dim \nu_\lambda < 1$, there is $\kappa > 0$ such that every convolution factor X of X_λ satisfies*

$$t \|f'_t\|_1 \geq \kappa \quad (t > 0), \quad (2.4)$$

where f_t is the density of $X + tZ$. We may take $\kappa \leq 1$.

Proof. First let $X = X_\lambda$, and write $X = \varepsilon + \lambda X'$. Exact dimensionality [FH09], compact support, and (2.3) give $h(X + tZ) = (1 - \alpha) \log t + o(\log(1/t))$ as $t \downarrow 0$. Writing $h_t = h(X + tZ)$, conditioning on ε gives

$$I_t := I(\varepsilon; X + tZ) = \chi + h_t - h_{t/\lambda}.$$

The left side increases as $t \downarrow 0$. Averaging this identity over $t = \lambda, \lambda^2, \dots, \lambda^N$ telescopes, so $I_t \leq \alpha\chi$ for every $t > 0$. De Bruijn's identity and monotonicity of Fisher information [Sta59] yield

$$(1 - \alpha)\chi \leq \int_t^{t/\lambda} u J(f_u) du \leq \frac{1}{2} t^2 (\lambda^{-2} - 1) J(f_t).$$

Thus $t^2 J(f_t) \geq c_0 := 2(1 - \alpha)\chi/(\lambda^{-2} - 1)$. The Gaussian score identity $f'_t(X + tZ)/f_t(X + tZ) = -t^{-1}\mathbb{E}[Z \mid X + tZ]$ gives a fourth moment at most $3t^{-4}$. Interpolation implies $t \|f'_t\|_1 \geq c_0^{3/2}/\sqrt{3}$. Finally, if $X_\lambda = X + Y$ independently, then the smoothed derivative of X_λ is $f'_t * \mathcal{L}(Y)$. Convolution decreases its L^1 norm. $\square$

3. ORDERED DETECTION AND SHIFTED GRIDS

For a bounded law ρ , write $I(a, r) = [a, a + r]$, let I_0, I_1 be its two halves, and let m_I be its conditional mean. Define

$$B_\rho(r) = \frac{1}{r} \int_{\mathbb{R}} \frac{\rho(I_0)\rho(I_1)}{\rho(I)} da, \quad (3.1)$$

$$V_\rho(r) = \frac{1}{r^3} \int_{\mathbb{R}} \int_{I(a, r)} (x - m_I)^2 d\rho(x) da, \quad (3.2)$$

$$v_\rho(r) = V_\rho(r) - \frac{1}{4} V_\rho(r/2). \quad (3.3)$$

An integrand with $\rho(I) = 0$ is zero. These are averages over grid origins. In particular $0 \leq B_\rho(r), V_\rho(r) \leq 1/4$. Total variance shows that $v_\rho(r) \geq 0$: it is the averaged variance of the two child means, normalized by r^2 . We use the same notation for a random variable with law ρ .

Lemma 3.1 (Ordered detection). *Suppose (2.4) holds. Let $U \in [0, r]$ be independent of X , and let $L \in \{0, 1\}$ distinguish two ordered intervals containing U . At $t = 4r/\kappa$,*

$$I(L; X + U + tZ) \geq \frac{\kappa^4}{128r^2} \text{Var}(\mathbb{E}[U \mid L]).$$

Proof. Put $p = \mathbb{P}(L = 1)$, $q = 1 - p$, and $m_i = \mathbb{E}[U \mid L = i]$. Couple the conditional laws as $U_0 \leq U_1$, and let g_i be their output densities. Since $\|f''_t\|_1 \leq 2t^{-2}$,

$$g_1 - g_0 = -(m_1 - m_0)f'_t + R, \quad \|R\|_1 \leq \frac{2r}{t^2}(m_1 - m_0).$$

This follows by integrating $f'_t(\cdot - s) - f'_t$ from U_0 to U_1 . Our choice of t gives $\|g_1 - g_0\|_1 \geq \kappa^2(m_1 - m_0)/(8r)$. Pinsker's inequality (see [CT06, Lemma 11.6.1]) for the mixture $qg_0 + pg_1$ gives $I(L; X + U + tZ) \geq pq \|g_1 - g_0\|_1^2/2$. The cases $pq = 0$ are immediate. $\square$

The next comparison is what prevents grid boundaries from losing entropy at many scales.

Lemma 3.2 (Shifted-grid comparison). *For every bounded law ρ ,*

$$B_\rho(r) \leq 64V_\rho(r) + 8 \sum_{\ell \geq 2} 2^{-\ell} V_\rho(2^{-\ell} r). \quad (3.4)$$

Consequently, with $r_j = 2^{-j} r$,

$$\sum_{j < k} B_\rho(r_j) \leq 68 \sum_{j < k} V_\rho(r_j) + 4, \quad (3.5)$$

$$\sum_{j < k} V_\rho(r_j) \leq \frac{4}{3} \sum_{j < k} v_\rho(r_j) + \frac{1}{12}. \quad (3.6)$$

Proof. Use pairs $x < y$, and put $\delta = y - x$. The pair kernels in $B_\rho(r)$ and $V_\rho(s)$ are respectively

$$\frac{1}{r} \int_{\{a: x \in I_0, y \in I_1\}} \frac{da}{\rho(I(a, r))}, \quad \frac{\delta^2}{s^3} \int_{y-s}^x \frac{db}{\rho(I(b, s))}.$$

For $\delta \geq r/8$ the first is at most 64 times the $V_\rho(r)$ kernel. For $0 < \delta < r/8$, choose the dyadic $s = 2^{-\ell} r$, $\ell \geq 2$, with $s/4 \leq \delta < s/2$. Every length- s interval containing x, y lies inside every length- r interval whose midpoint separates them: here $s \leq r/2$. The corresponding sets of starting points have lengths $s - \delta$ and δ . Thus the first kernel is at most

$$\frac{\delta}{r(s - \delta)} \int_{y-s}^x \frac{db}{\rho(I(b, s))} \leq \frac{8s}{r} \frac{\delta^2}{s^3} \int_{y-s}^x \frac{db}{\rho(I(b, s))}.$$

Integrate over pairs and sum over s . This proves (3.4). Summing over j and using $V \leq 1/4$ beyond the last scale gives (3.5). Finally,

$$\sum_{j < k} v_\rho(r_j) = \frac{3}{4} \sum_{j < k} V_\rho(r_j) + \frac{1}{4} V_\rho(r) - \frac{1}{4} V_\rho(r_k),$$

which proves (3.6). $\square$

Lemma 3.3 (Entropy supplies variance). *There are absolute $c, C > 0$ with the following property. If $a < b$, $\ell = b - a \geq 1$, and*

$$\overline{H}(Y; e^{-b}) - \overline{H}(Y; e^{-a}) \geq H > 0,$$

then

$$\int_a^b v_Y(e^{-s}) ds \geq \frac{cH}{\log(e + \ell/H)} - C. \quad (3.7)$$

Proof. For a binary probability p and $z = p(1 - p)$, $h_2(p) \leq 2z \log(e/z)$. The function $z \log(e/z)$ is concave. For a shifted nested grid with k binary refinements, the entropy chain rule and Jensen therefore give

$$\overline{H}(Y; 2^{-k} r) - \overline{H}(Y; r) \leq 2S \log(ek/S), \quad S = \sum_{j < k} B_Y(2^{-j} r).$$

If this entropy difference is at least $H_0 > 0$, inversion of $z \log(e/z)$ gives $S \geq cH_0 / \log(e + k/H_0)$, after an absolute change of c . One may check the inversion directly from $\phi(u/[2 \log(e/u)]) \leq u$ for $\phi(z) = z \log(e/z)$ and $0 < u \leq 1$. Equations (3.5)–(3.6) give the same lower bound, minus an absolute constant, for $\sum_{j < k} v_Y(2^{-j} r)$.

Set $k = \lfloor \ell / \log 2 \rfloor - 1$ and average over $r = e^{-(a+u)}$, $0 \leq u \leq \log 2$. Each endpoint of the resulting entropy interval lies within two binary scales of the desired endpoint.

The entropy loss is bounded by an absolute constant by the grid comparison following (2.3). The averaged sums of v integrate over a subinterval of $[a, b]$. Since $v \geq 0$, this proves (3.7). Bounded ℓ or H are absorbed in C . $\square$

Lemma 3.4 (Localized information gain). *Suppose (2.4) holds and X, Y are independent and bounded. For $0 < t < R$ and $P \geq 0$ with $U := Re^{-P} \geq t$,*

$$D_{X,Y}(t, R) \geq c_\kappa \int_t^U v_Y(r) \frac{dr}{r} - \frac{e^{-2P}}{8}, \quad c_\kappa = \frac{\kappa^4}{128 \log 2}. \quad (3.8)$$

Proof. Fix $u \in [U/2, U]$ and a grid origin $\theta \in [0, u)$. Let $K_j = \lfloor 2^j(Y + \theta)/u \rfloor$. In (2.2), the difference of the K_0 -information terms is nonnegative. In each K_0 -cell the input has variance at most $u^2/4$, so Gaussian capacity and data processing give

$$I(Y; X + Y + RZ \mid K_0) \leq u^2/(8R^2).$$

Writing $O_t = X + Y + tZ$, the finite information chain gives

$$I(Y; O_t \mid K_0) \geq \sum_{j=0}^J I(K_{j+1}; O_t \mid K_j).$$

Take precisely the parents with mesh $r_j = 2^{-j}u \geq t$. For each term, separately degrade the output noise from t to $4r_j/\kappa$. Lemma 3.1, averaged over θ , gives

$$D_{X,Y}(t, R) \geq \frac{\kappa^4}{128} \sum_{\{j \geq 0: 2^{-j}u \geq t\}} v_Y(2^{-j}u) - \frac{e^{-2P}}{8}.$$

The law of X stays unchanged when a Y -label is conditioned on. If $u < t$ the sum is empty and the inequality still holds by (2.2). Finally average u with measure $du/(u \log 2)$ on $[U/2, U]$. The dyadic intervals partition $(0, U]$, so the averaged sum is exactly $(\log 2)^{-1} \int_t^U v_Y(r) dr/r$. $\square$

4. PACKING INDEPENDENT BLOCKS

Fix $\lambda \in (1/2, 1)$ with $\dim \nu_\lambda < 1$, and choose

$$\chi < b < h < \log 2. \quad (4.1)$$

All constants in this section may depend on these choices and on the roughness constant κ . Since the support of B_n lies in $[0, (1 - \lambda)^{-1}]$,

$$\overline{H}(B_n; e^{-bn}) \leq bn + O_\lambda(1). \quad (4.2)$$

Suppose, for a length n and a number $F > b$, that

$$\overline{H}(B_n; e^{-Fn}) \geq hn. \quad (4.3)$$

Lemma 3.3 gives

$$\int_{bn}^{Fn} v_{B_n}(e^{-s}) ds \geq \frac{cn}{\log(e + F)} \quad (4.4)$$

whenever $n/\log(e + F)$ is sufficiently large.

Proposition 4.1 (Packing bound). *Let (n_j, F_j) , $1 \leq j \leq J$, satisfy (4.3)–(4.4). Put $L = (b - \chi)n_1/4$ and suppose*

$$(b - \chi)n_{j+1} > F_j n_j + L + 4\chi. \quad (4.5)$$

There is a constant C_0 such that, if

$$\log \left(2 + C_0 \sum_{j=1}^J (F_j + 1) \right) \leq L/2, \quad L \geq 4\chi + 2,$$

then

$$\sum_{j=1}^J \frac{1}{\log(e + F_j)} \leq C_0. \quad (4.6)$$

Proof. Let $N_* = C_1 \sum_j (F_j + 1)$ be an upper bound for the number of blocks selected below, with C_1 sufficiently large, and set $P = \log(2 + N_*)$. Choose T so large that all coefficient indices below are nonnegative, and put $R = e^{-T}$, $t = e^{-(T+L)}$. The inner logarithmic window is $W = [T + P, T + L]$ and has length $L - P \geq L/2$.

For a fixed j , use starts $m \in \mathbb{Z}_{\geq 0}$ with

$$m\chi \in [T + P - F_j n_j, T + L - b n_j]. \quad (4.7)$$

Select those starts in one residue class modulo n_j . The blocks

$$Y_m = \sum_{i=m}^{m+n_j-1} \varepsilon_i \lambda^i \stackrel{\mathcal{L}}{=} \lambda^m B_{n_j}$$

are disjoint within this family. Averaging over the n_j residue classes, and using $v_{\lambda^m B_{n_j}}(e^{-s}) = v_{B_{n_j}}(e^{-(s-m\chi)})$, gives

$$\frac{1}{n_j} \sum_{\text{all starts } m} \int_W v_{\lambda^m B_{n_j}}(e^{-s}) ds \geq \frac{L-P}{2\chi n_j} \int_{bn_j}^{F_j n_j} v_{B_{n_j}}(e^{-u}) du.$$

Indeed, for each u in the latter interval, at least $\lfloor (L-P)/\chi \rfloor \geq (L-P)/(2\chi)$ integers m satisfy $u + m\chi \in W$, and all satisfy (4.7). Choose a residue class attaining the average. By (4.4), its contribution is at least $c(L-P)/\log(e + F_j)$.

There are at most $C_1(F_j + 1)$ selected blocks in family j . All their coefficient indices lie between

$$(T + P - F_j n_j)/\chi - 1 \quad \text{and} \quad (T + L - (b - \chi)n_j)/\chi + 1.$$

Condition (4.5) makes these ranges disjoint for successive families, with family $j+1$ lying before family j . Thus all selected blocks are independent.

Start with the sum of the unused digits and add the selected blocks one at a time. Every intermediate background is a convolution factor of X_λ , hence satisfies Lemma 2.1 with the same κ . Apply Lemma 3.4 to each addition and telescope. By (2.1), the total gain is at most L . It follows that

$$L \geq c(L-P) \sum_{j=1}^J \frac{1}{\log(e + F_j)} - \frac{N_* e^{-2P}}{8}.$$

The last term has absolute value at most 1, and $L - P \geq L/2$. Increasing C_0 proves the claim. $\square$

Corollary 4.2. *The assertions (1.1) and (1.2) hold for every $\chi < h < \log 2$.*

Proof. For fixed $F > b$, suppose (4.3) holds at infinitely many lengths. Choose arbitrarily many of them successively far enough apart to satisfy (4.5). For any prescribed number J , the first length can be taken large enough that all hypotheses

of Proposition 4.1 hold. The bound $J/\log(e + F) \leq C_0$ is impossible for large J . For a smaller fixed exponent, comparison with any fixed $F > \max(b, C)$ gives the assertion, with a bounded grid error absorbed by first choosing a slightly smaller h . Indeed, a fine cell meets at most two coarse cells.

Now fix $A > 0$, and suppose (4.3) holds for all large n with $F_n = A \log n$. Choose n_1 large and put

$$J = \lceil (\log \log n_1)^2 \rceil, \quad n_{j+1} = \lceil D(A \log n_j + 1)n_j \rceil,$$

where D is a fixed sufficiently large constant. The separation condition holds. Moreover $n_j \leq n_1^2$ for $j \leq J$ when n_1 is large: the logarithm increases by $O(\log \log n_1)$ at each step, and $(\log \log n_1)^3 = o(\log n_1)$. Thus the number of blocks is $O(J \log n_1)$ and its logarithm is $O(\log \log n_1) = o(L)$. Proposition 4.1 applies, whereas

$$\sum_{j=1}^J \frac{1}{\log(e + A \log n_j)} \geq \frac{J}{\log(e + 2A \log n_1)} \rightarrow \infty.$$

This is a contradiction. $\square$

Corollary 4.3. *For every $C > 0$ and every sufficiently large n , there is a nonzero polynomial P of degree less than n , with coefficients in $\{-1, 0, 1\}$, such that $|P(\lambda)| < e^{-Cn}$.*

Proof. Apply (1.1) with $h < \log 2$. For some grid origin the entropy of the length- n word label is smaller than $n \log 2$. Two distinct words therefore occupy one grid cell. Their difference is the required polynomial. $\square$

5. THE ARITHMETIC INGREDIENTS

Let $\mathcal{P}_n$ denote the nonzero polynomials of degree at most n with coefficients in $\{-1, 0, 1\}$. We fix a compact interval $I \subset (0, 1)$. For an algebraic number η , write $M(\eta)$ for its Mahler measure and $d(\eta)$ for its degree.

Lemma 5.1. *There is an integer $q = q_I \geq 1$ such that every $P \in \mathcal{P}_n$, after removal of its initial power of x , has at most q zeros, counted with multiplicity, in a fixed disk containing I and contained in the unit disk. There are also $c_I > 0$ and $A_I < \infty$ such that, for n large, any nonempty family $\mathcal{A} \subset \mathcal{P}_n$ satisfying*

$$|P(\lambda)| \leq r = e^{-A_n \log n} \quad (P \in \mathcal{A}), \quad \lambda \in I, \quad A \geq A_I,$$

has a common real zero η with $|\eta - \lambda| \leq r^{c_I}$.

Proof. Choose $\sup I < R_0 < R_1 < 1$. A normalized polynomial Q has $|Q(0)| = 1$ and $|Q(z)| \leq (1 - R_1)^{-1}$ on $|z| = R_1$. Jensen's formula bounds the number of its zeros in $|z| \leq R_0$ by a constant q_I . In the product formula for $Q(\lambda)$, all other roots are a fixed positive distance from I . Removing a power of x costs at most an exponential factor in n . Consequently a value at most r forces a root within $(e^{C_I n r})^{1/q_I} \leq r^{c_I}$, with a fixed $c_I > 0$.

We use Mahler's classical separation inequality [Mah64]: a squarefree integer polynomial of degree d has distinct roots separated by at least $\sqrt{3} d^{-(d+2)/2} M(P)^{-(d-1)}$. Two distinct roots of members of $\mathcal{P}_n$ are therefore separated by $e^{-C_n \log(n+1)}$. To see this, take their minimal polynomial if they are conjugate, or the product of their minimal polynomials otherwise; its degree is at most $2n$ and its Mahler measure is at most $n + 1$, since $M(P) \leq \|P\|_2 \leq \sqrt{n+1}$. For A_I sufficiently large, all the

nearby roots found above must thus be the same. Complex conjugation shows this root is real. $\square$

This common-root argument is the root-separation alternative explicitly noted in [BV19, Section 3].

Lemma 5.2 (Entropy transfer). *Fix $\lambda \in I$. If A and then n are sufficiently large and*

$$\overline{H}(B_n(\lambda); e^{-An \log n}) < hn < n \log 2,$$

then there is a real algebraic unit η , of degree at most n , such that

$$|\eta - \lambda| \leq e^{-c_I An \log n}, \quad h_\eta := \lim_{m \rightarrow \infty} \frac{H(B_m(\eta))}{m} < h.$$

Proof. Choose a grid origin whose entropy is at most the average, and apply Lemma 5.1 to every nonzero difference of words in a common cell. The family is nonempty because $h < \log 2$. All words in each cell have the same value at η . Hence $H(B_n(\eta)) \leq \overline{H}(B_n(\lambda); e^{-An \log n})$. Subadditivity gives the entropy-rate bound. A nonzero root of a $\{-1, 0, 1\}$ polynomial is an algebraic unit: remove its initial power of x and note that both end coefficients have modulus one. $\square$

Lemma 5.3 (Breuillard–Varjú [BV20], [Var19a]). *For every $h < \log 2$ there is $M_h < \infty$ such that $h_\eta < h$ for a real algebraic unit $\eta \in (0, 1)$ implies $M(\eta) \leq M_h$.*

Proof. We recall the Gaussian argument, to keep this input explicit. There is a real contraction matrix A whose eigenvalues are the conjugates of η inside the unit disk and whose singular values are $1, \dots, 1, M(\eta)^{-1}$. Indeed, start with a real diagonalizable matrix A_0 with that spectrum and a cyclic vector w for A_0^t . The positive quadratic form

$$Q(x) = \sum_{j \geq 0} \langle A_0^j x, w \rangle^2$$

satisfies $Q(x) - Q(A_0 x) = \langle x, w \rangle^2$. In Q -orthonormal coordinates all but one singular values are one; the determinant determines the last.

For this matrix put $E_A(Y) = h(Y + AZ) - h(Y + Z) - \log |\det A|$. Here Z is a standard Gaussian vector in the matrix dimension. Gaussian degradation shows that E_A increases on adding an independent summand. If $X_A = \sum_{j \geq 0} \varepsilon_j A^j x$, its self-similarity gives

$$nE_A(X_A) \leq E_{A^n}(X_A) \leq H\left(\sum_{j < n} \varepsilon_j A^j x\right).$$

For the first inequality, telescope and use that $A^{-j} X_A$ is a sum of an independent copy of X_A and a finite prefix. The second inequality conditions on the first n digits. Polynomial relations at η hold at A , so

$$h_\eta \geq E_A(X_A) \geq E_A(\varepsilon_0 x).$$

Write $M = M(\eta)$, choose a unit vector v with $\|A^{-1}v\| = M$, and set $x = M^{-1/2}v$. Whitening the Gaussian gives

$$h_\eta \geq I(\varepsilon_0; \sqrt{M} \varepsilon_0 + Z) - I(\varepsilon_0; M^{-1/2} \varepsilon_0 + Z).$$

The first term tends to $\log 2$ by binary decoding, and the second is at most $1/(8M)$ by Gaussian capacity. This proves the assertion. $\square$

Lemma 5.4 (An arithmetic annulus; cf. Dimitrov's argument in [RV24, Lemma 4.6]). *Fix $I \subset (0, 1)$ compact and $M \geq 1$. There are constants $a, B, D > 0$ such that the following holds for all sufficiently large N . Suppose $\eta, \lambda \in I$, η is algebraic of degree d , $M(\eta) \leq M$,*

$$N \geq Dd \log(N+1), \quad e^{-aN} \leq |\lambda - \eta| \leq e^{-a(N-1)}. \quad (5.1)$$

Then every $P \in \mathcal{P}_N$ satisfies

$$|P(\lambda)| \geq e^{-BN}. \quad (5.2)$$

Proof. First remove any initial power of x , so $|P(0)| = 1$. Let k be the least order with $P^{(k)}(\eta) \neq 0$. Lemma 5.1 gives $0 \leq k \leq q_I$. The integer polynomial $P^{(k)}$ has coefficient height at most $(N+1)^{q_I}$. The resultant with the primitive minimal polynomial of η is a nonzero integer. Bounding the values at the other conjugates by coefficient height times the appropriate power of $\max(1, |z|)$ gives

$$|P^{(k)}(\eta)| \geq M^{-N} (N+1)^{-(q_I+1)(d-1)}.$$

Choosing D large enough, and then N large, ensures

$$|P^{(k)}(\eta)|/k! \geq e^{-(\log M+1)N}.$$

Every derivative of bounded order of a normalized bounded-coefficient polynomial is uniformly bounded on a fixed neighborhood of I inside $(-1, 1)$. Taylor's theorem, with $\delta = \lambda - \eta$, therefore gives

$$P(\lambda) = \frac{P^{(k)}(\eta)}{k!} \delta^k + O_I(|\delta|^{k+1}).$$

Take $a = \log M + 3$. By (5.1), the remainder is at most half the main term for large N . Thus

$$|P(\lambda)| \geq \frac{1}{2} e^{-(\log M+1)N} e^{-aq_I N}.$$

Restoring a removed power of x costs at most $(\min I)^N$. Increasing B proves the lemma. $\square$

6. COMPLETION AND THE SCOPE OF THE CLAIM

Proposed proof of Theorem 1.1. Suppose λ is transcendental and $\dim \nu_\lambda < 1$. Fix $\chi < h < \log 2$ and a compact interval $I \subset (0, 1)$ containing λ in its interior. Let $M = M_h$ be given by Lemma 5.3, and let a, B, D be the constants in Lemma 5.4. Choose A large enough for Lemma 5.2 and so that $c_I A > 4aD$.

By (1.2) and Lemma 5.2, for arbitrarily large n there are algebraic units η with

$$d(\eta) \leq n, \quad M(\eta) \leq M, \quad 0 < \delta := |\eta - \lambda| \leq e^{-c_I A n \log n}.$$

The strict inequality uses transcendence. For large n we have $\eta \in I$. Choose $N = \lceil a^{-1} \log(1/\delta) \rceil$. Then the annulus condition holds and $N \geq (c_I A/a) n \log n$. Since $x/\log(x+1)$ is increasing for large x , the choice of A implies $N \geq Dd(\eta) \log(N+1)$. Lemma 5.4 now bounds every member of $\mathcal{P}_N$ below by e^{-BN} at λ . Corollary 4.3, applied with exponent $B+1$, supplies a member of $\mathcal{P}_{N-1}$ violating this bound. This contradiction proves the theorem. $\square$

The different quantifiers are essential: low entropy at $e^{-An \log n}$ is needed only along a subsequence, whereas small polynomial values at fixed exponential precision are needed at every large degree. A companion audit records the scale and conditioning checks. The localized estimate and the packing argument remain the main priorities for independent review.

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